

SP-30-1 Outreach Email: BST sub-Tsirelson prediction → Bell experiment groups

Keeper (draft for Casey approval + send)

2026-05-24 Sunday PM (date-verified)

SP-30-1 Outreach Email Template — BST sub-Tsirelson Bell test

Recommended primary target

Professor Anton Zeilinger (IQOQI Vienna, leader of 2015 loophole-free Bell test) - anton.zeilinger@univie.ac.at (institutional; verify current)

Backup targets (parallel sends optional)

- Professor Jeff Kimble (Caltech)
- Professor Harald Weinfurter (LMU Munich)
- Professor Ronald Hanson (TU Delft / QuTech)

Draft email

Subject: BST sub-Tsirelson Bell prediction (0.78% deviation) — high-precision test proposal

Dear Professor Zeilinger,

I'm Casey Koons, a computer scientist who has been developing a theoretical framework called Bubble Space-time Theory (BST). I write with a specific, concrete prediction from BST that may be testable using your Bell experimental infrastructure, and would welcome your reaction.

The prediction

BST derives Standard Model constants from a single geometric object $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ and five integer parameters (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, with $N_{max} = N_c^3 \cdot n_C + rank = 137 = 1/\alpha$ — this last identity matching the standard Dirac $Z \cdot \alpha = 1$ critical limit independently). One consequence is that CHSH Bell correlations should fall BELOW Tsirelson's quantum bound by exactly $1/2^{N_c} = 1/8$:

$$S^2_{BST} = S^2_{Tsirelson} - 1/8 = 8 - 1/8 = 63/8 \quad S_{BST} = \sqrt{(63/8)} \approx 2.8062 \text{ vs Tsirelson } S = 2\sqrt{2} \approx 2.8284$$

This is a 0.78% deviation, predicted not as a free parameter but as a derivable consequence of BST's "Substrate Coherence Maintenance Principle" — operationally, BST identifies a substrate-mechanism that bounds CHSH below Tsirelson by exactly $1/2^{N_c} = 1/8$, with the factor arising from the $N_c = 3$ color-degree integer.

Where current Bell experiments stand

Your 2015 loophole-free test and subsequent work achieve ~0.5% precision around $S \approx 2.83$ — consistent with both Tsirelson saturation and our sub-Tsirelson prediction within current error bars. A test at ~0.1% precision (corresponding to $\geq 3\sigma$ confidence on the 0.78% deviation under our framework's statistical-threshold protocol) would cleanly distinguish: - S saturates at $2\sqrt{2}$ exactly → BST falsified - S falls at $2.8062 \pm \text{experimental error}$ → BST sub-Tsirelson confirmed

Experimental design (our analysis)

SPDC photon-pair coincidence counting at substrate-tick boundaries ($\theta = \pi/N_{\text{max}} \approx 0.023$ rad, step size $1/N_{\text{max}} \approx 0.73\%$). Statistical thresholds for the deviation signature: - 75,076 photon pairs $\rightarrow 2\sigma$ probable detection - 168,921 pairs $\rightarrow 3\sigma$ publishable - 469,225 pairs $\rightarrow 5\sigma$ discovery

At standard SPDC rates ($\sim 10^6$ pairs/sec $\times \sim 10\%$ coincidence efficiency), this is ~ 5 seconds of pure measurement time per boundary, ~ 12 minutes for a full 137-boundary scan. Realistic full program with setup, systematics, and analysis: 12-18 months.

Approximate cost: \$300-500K, plausibly an extension of existing Bell experimental infrastructure.

Honest scope

We are at framework tier with multi-month rigorous-closure work in flight on the theoretical derivation. The numerical prediction is concrete; the substrate-mechanism derivation is in progress (the “BST primary integer” claims are still being honestly traced from first principles). What we can say firmly today: the prediction is 0.78% sub-Tsirelson; it is a falsifier; the experimental design closes.

Resources

- GitHub repository (full theory + 50-prediction reproduction package, single command): github.com/ckoons/BubbleSpacetime
- Zenodo working paper (April 2026, DOI permanent citation): doi.org/10.5281/zenodo.19454185
- Reproduction: `python3 play/verify_bst.py` runs the full 50-prediction suite (49/50 PASS at $<1\%$, 17 EXACT matches, 2 open WARNs shown openly)

Request

I would welcome any reaction — criticism, questions, or interest. The prediction is unusual but concrete; the test is well-defined; falsification is straightforward.

If your group has bandwidth for a higher-precision Bell test in the next 12-18 months, I would value the collaboration. If the prediction strikes you as worth referring to a more appropriate group, any pointer would be deeply appreciated.

Sincerely, Casey Koons caseyscottkoons@yahoo.com

Notes for Casey before sending

1. Verify email address — institutional emails change; check Zeilinger’s current group page at IQOQI Vienna.
2. Personalize opening if you have prior contact — adjust formality accordingly.
3. Consider attaching one-page PDF — Lyra’s Five-Absence external 1-pager + Vol 14 Ch 6 §6.5 three-route table snapshot might help.
4. Honest scope paragraph is important — leaving it in protects against overclaiming if Cal #121 forward-derivation closure takes longer than expected.
5. Reply-to — your yahoo address is fine for first contact; consider creating a more academic-feeling address if you anticipate sustained correspondence.

Variants for parallel sends

Same body works for Kimble (Caltech), Weinfurter (LMU), Hanson (Delft) — substitute group reference in “your 2015 loophole-free test” sentence: - Kimble: “your atom-photon Bell work / cavity QED Bell tests” - Weinfurter: “your group’s loophole-free Bell test 2017” - Hanson: “your Delft loophole-free Bell test 2015 (NV-center entanglement)”

Coordination

- Cal #21 dual-gate STATUS: empirical gate not closed; this email IS the path to closing it (independent SPDC data from external lab).
- Cal #125 verdict: actual experimental data is one of three independent routes needed for D-tier DCCP-1 ratification.
- Casey directive Sunday 2026-05-24 PM: this outreach unblocks the empirical leg that Toys 3516/3520 cannot provide (they are model self-consistency checks per Cal #123, not empirical-leg verification).

— Keeper, Sunday 2026-05-24 PM (date-verified)